

The Barbelo Prime Tree

A New Visual Representation of Prime Gap Structure

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Abstract

We present a novel visual representation of the natural numbers in which prime numbers form a vertical *trunk*, and the composite numbers between each pair of consecutive primes are placed as a *branch* — alternating left and right with each new prime gap. The resulting structure resembles a tree, and makes prime gap sizes immediately visible without counting. We describe two variants of the construction, discuss their visual properties, and observe that the tree structure remains stable up to at least 10,000.

1. Introduction

Prime numbers have been represented and studied in many ways: the Sieve of Eratosthenes, the Ulam spiral, the prime number line, and various statistical plots of gap distributions. Each representation reveals different aspects of prime structure.

The construction presented here arose from an intuitive question: *what if the gaps between primes were made visible as physical space, rather than as a counted distance?* The original idea emerged from hand-drawn Excel worksheets in which primes were written in a central column and non-primes were placed beside them. What appeared was a tree-like form — narrow at the root, widening as gaps grow, but never losing its recognizable shape.

We call this the Barbelo Prime Tree, named after its originator.

2. Construction

The construction rule is as follows:

Trunk: Write all prime numbers vertically in a central column, one per row.

Branches: The composite numbers between two consecutive primes p_n and p_{n+1} are placed as a group — all to the right if the previous group was placed to the left, and vice versa. The side alternates with each prime gap.

Branch width: The width of each branch equals the prime gap minus one, i.e. the count of composite numbers between two consecutive primes.

This gives an unambiguous, deterministic layout for every natural number greater than 1.

3. Two Variants

Variant A — Horizontal branches (original): Branches extend horizontally left or right from the trunk. The gap size is visible as the horizontal extent of the branch. The trunk itself shows only primes, making the primality of each trunk entry immediately obvious. Large gaps produce wide, asymmetric arms on one side.

Variant B — Diagonal branches (developed variant): Branches extend diagonally downward from the trunk, at an angle. This has a key additional property: the gap is visible not only in the branch width, but also as a vertical displacement in the trunk — the next

prime "arrives later" when the gap is large. The overall shape resembles a fishbone or herringbone pattern.

Variant B encodes gap information twice — in branch length and in trunk rhythm — making it arguably more information-dense. Variant A is visually cleaner and more immediately tree-like.

4. Observations

Visual inspection of the tree up to $N = 10,000$ (containing 1,229 primes) reveals the following:

Stability: The tree form remains recognizable throughout the entire range. It does not collapse into noise or become unreadably dense.

Twin primes: Pairs of consecutive primes with gap 2 (twin primes) produce single-cell branches — the minimum possible branch width. These are visible as thin notches along the trunk.

Large gaps: The largest gap in the range 2–10,000 is 36 (between 9,551 and 9,587). This produces the widest branch, clearly visible as an outlier in the tree form.

Asymmetry: Because branches alternate sides, the tree is locally asymmetric but globally balanced. No systematic left or right bias is observed.

5. Open Questions

This representation is primarily visual and descriptive. We leave the following questions open for further investigation:

Does the maximum branch width grow without bound as N increases, and if so, at what rate? (This is equivalent to asking whether prime gaps are unbounded, which is known to be true.)

Is the tree form self-similar in any measurable sense? The visual impression of stability across scales suggests possible self-affine structure, but this has not been quantified.

Can the alternating left-right assignment of branches be used to define a useful integer sequence or encoding?

6. Implementation

An interactive implementation is available as a standalone HTML file. It uses the Sieve of Eratosthenes to compute primes up to a user-selected limit (up to 10,000), renders the tree on an HTML5 Canvas, and includes a secondary chart of the gap sequence. The implementation is available at: <https://www.insleben.com/Prime-Tree/primzahlenbaum.html>

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